

GEOMETRIC FLOWS (101)

[Bibliography?]

very imprecise

- Definitions in Mathematics:

- abstract $\leftrightarrow$ concrete

capture an intuitive notion, fit a result, ...

old/new maths: shift of focus? intention & systematisation?

- relations & classification

ordering possible instances to simplify (equivalence)

adjust to the existing framework & enrich it

$\rightarrow$ Geometry / Topology: "quantitative / qualitative closeness"

Classification results and interrelation ("best metric")

Uniformisation theorem (Klein - Koebe - Poincaré ~ 1900)

Every 1-conn. 1-dim. complex manifold is conformally

conn. + $\pi_1 \cong 0$

equivalent to $\mathbb{D}^2, \mathbb{C}, \mathbb{S}^2$

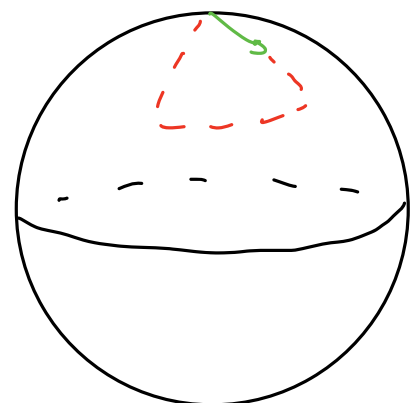
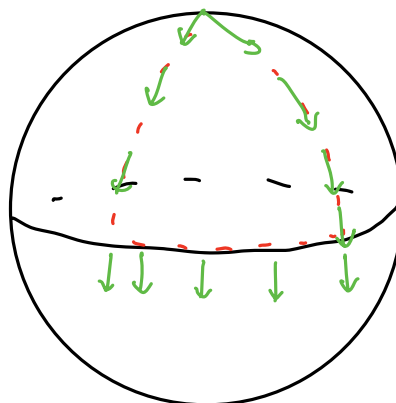
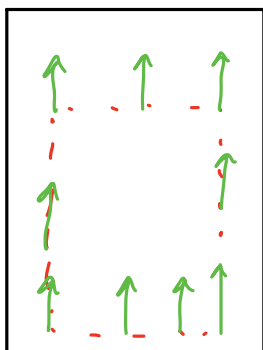
$K = -1, 0, 1$

$$[g] = [h]$$

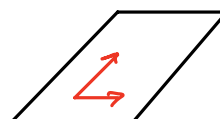
$$\Leftrightarrow g = \psi h \text{ w/ } \psi > 0$$

"angle preserving"

Curvature

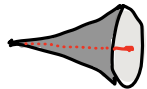
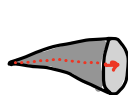
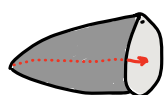
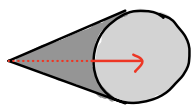


do for every 2-plane

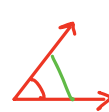


$R_{ijkl}(p)$

"Averages" : Ricci $\xrightarrow{\quad\quad\quad}$ Sectional
 (traces) $\text{Ric}_{ij}(p)$ $\text{Sec}(p)$



$K > 0$



$K = 0$



$K < 0$

$\rightarrow$ push-fwd. metric : $\tilde{\Sigma} \rightarrow \Sigma$
 $\approx$ natural univ. cover (Riemannian)

constant curvature metric in each conformal class



Nobody :

Mathematicians: GENERALISE

- Kähler-Einstein metrics

- Yamabe problem

$(M^{n \geq 3}, g)$ compact, $\partial M = \emptyset \Rightarrow \exists \tilde{g}$ s.t. $\text{sec}_{\tilde{g}} = \text{const.}$ & $[\tilde{g}] = [g]$

- Compact w/ boundary, non-compact ?

$\hookrightarrow$ boundary conditions (mean curvature, ∞ , completeness, ...)

Reformulation: $\tilde{g} = u^{\frac{4}{n-2}} g$ (u positive function on the manifold)

Calculating relationship between sectional curvatures:

$$\text{sec}_{\tilde{g}} u^{\frac{n+2}{n-2}} = \text{sec}_g u - \frac{4(n-1)}{n-2} \Delta_g u$$

Impose $\text{sec}_{\tilde{g}} = c$ constant $\leadsto$ equation for u :

$$c u^{\frac{n+2}{n-2}} - \text{sec}_g u + \frac{4(n-1)}{n-2} \Delta_g u = 0$$

Principle of "least" action: (stationary)



- Fermat: light travels following the shortest path
(refraction)
- Maupertuis / Euler: path taken by a system to evolve from an initial to a final state is the one for which the action is stationary ($\approx$ critical point, many cases a minimum)

Action: functional, "energy": yielded by math. description of the system
Useful to compare field theories. Normally integral on $\int_{t_1}^{t_2} \mathcal{L} dt$
 $\uparrow$ Lagrangian

Einstein-Hilbert action & GR

$$S(g) \propto \int (\text{scal}_g + \underbrace{\mathcal{L}_{\text{matter}}}_{\text{can be omitted}} - \underbrace{\Lambda}_{\text{can be omitted}}) \text{dvol}_g$$

$$\delta S = 0 \leadsto g \text{ satisfies } \text{Ric} - \frac{1}{2} g \text{ scal}_g + \underbrace{\Lambda g}_{\text{can be omitted}} = \underbrace{\frac{8\pi G}{c^4} T}_{\text{can be omitted}}$$

$$\text{Simplest: "Einstein manifold" } \text{Ric} = \frac{1}{2} g \text{ scal}_g$$

Equivalent to Yamabe problem: find critical point for Yamabe functional

$$E_g(u) = \frac{\int_M \left(\frac{4(n-1)}{n-2} |du|_g^2 + \text{scal}_g u^2 \right) \text{dvol}_g}{\left(\int_M u^{\frac{2n}{n-2}} \text{dvol}_g \right)^{\frac{n-2}{n}}} \approx S \left(\underbrace{u^{\frac{4}{n-2}} g}_{\tilde{g}} \right)$$

Idea ($\approx$ Neural Networks):

- (1) Start with initial config.
- (2) Evolve to better configuration (closest to critical point)
- (3) Converge to critical point

$$(1) \quad g(0) = g_0$$

(2) There is a standard formulation: "gradient flow"

$$\mathcal{E}(\phi) \text{ energy functional} \leadsto \frac{\partial \phi}{\partial t} = - \frac{\partial \mathcal{E}}{\partial \phi}$$

Yamabe PDE (Yamabe flow):

$$\frac{\partial g(t)}{\partial t} = - \left(\text{scal}_{g(t)} - \frac{\int_M \text{scal}_g \, d\text{vol}_g}{\text{vol}_g(M)} \right) g(t) \quad , \quad g(0) = g_0$$

(intrinsic) GEOMETRIC FLOW

mean scalar curvature $\equiv r_{g(t)}$

(3) Study convergence: expected that $g(t) \xrightarrow[t \nearrow \infty]{} \tilde{g}$ with $\text{scal}_{\tilde{g}} = c$

• Geometric flow:

- (Parabolic) PDE of the form $g_t = Lg$ w/ L (elliptic) operator
- This PDE is the gradient flow associated with a functional, which is usually related to the L^2 -norm of some curvature

Yamabe $\approx L^2$ -norm of scal_g + normalisation condition

Ricci $\approx \text{Ric}_g$

On another note ...

(for now, topological)

Next step to classif. of manifolds $\leadsto$ classif. of morphisms: $\text{Diff}(X)$
(up to diff.)

- Broad theory on $\text{MCG}(X) := \text{Diff}(X) / \text{Diff}_0(X)$ (Teichmüller theory, ...)
 $\leadsto$ focus on $\text{Diff}_0(X)$

- Smooth fibre bundles with fibre X have $\text{Diff}(X)$ as structure group

$O(2) \hookrightarrow \text{Diff}(S^1)$ deformation retract $(\text{Isom}(S^n) \cong O(n+1))$

(Smale) $O(3) \hookrightarrow \text{Diff}(S^2)$ homotopy equivalence

Proof: $\Leftrightarrow \text{Met}_{K=1}(S^2)$ contractible

$\text{Diff}(S^2) \curvearrowright \text{Met}_{K=1}(S^2)$ by pushfwd, transitive:

$$\text{Met}_{K=1}(S^2) \cong \text{Diff}(S^2) / \underline{\text{Isom}(S^2, g)}$$

stabiliser of any elem., e.g. round metric on S^2
 $\cong O(3)$

Fibre bundle: $O(3) \rightarrow \text{Diff}(S^2) \rightarrow \text{Met}_{K=1}(S^2)$

$\leadsto$ long exact sequence of homotopy groups

$$\begin{array}{l} \nearrow \pi_0(O(3)) \xrightarrow{\cong} \pi_0(\text{Diff}(S^2)) \longrightarrow \pi_0(\text{Met}_{K=1}(S^2)) \\ \nearrow \pi_1(O(3)) \xrightarrow{\cong} \pi_1(\text{Diff}(S^2)) \longrightarrow \pi_1(\text{Met}_{K=1}(S^2)) \\ \nearrow \pi_2(O(3)) \xrightarrow{\cong} \pi_2(\text{Diff}(S^2)) \longrightarrow \pi_2(\text{Met}_{K=1}(S^2)) \\ \vdots \end{array}$$

To show $\text{Met}_{K \equiv 1}(\mathbb{S}^2)$ contractible: Ricci flow

$$\frac{\partial g}{\partial t} = -2 \text{Ric}_g g, \quad g(0) = h$$

(Hamilton - Chow)

$(g(t))_{t \in [0, T]}$ Ricci flow on $\mathbb{S}^2$ (curvature blows up when $t \rightarrow T$)

$\Rightarrow T < \infty$ & (modulo rescaling) $g(t) \xrightarrow[t \rightarrow T]{} \tilde{g} \in \text{Met}_{K \equiv 1}(\mathbb{S}^2)$

$\Rightarrow \text{Met}_{K \equiv 1}(\mathbb{S}^2) \hookrightarrow \underline{\text{Met}(\mathbb{S}^2)}$ deformation retract
contractible (convex $\subseteq$ affine)

Other surfaces:

$g = 1 \quad \text{Diff}^0(X) \simeq \mathbb{T}^2$
+ use $\text{MCG}(X)$

$g \geq 2 \quad \text{Diff}^0(X)$ contractible

$O(4) \hookrightarrow \text{Diff}(\mathbb{S}^3)$? more complicated, shows $\text{Met}_{K \equiv 1}(\mathbb{S}^3)$ contractible
(Smale "conjecture")

Generalised Smale conj.: other 3-manifolds? (compact, conn.)

$\text{Isom}(X, g_X) \hookrightarrow \text{Diff}(X)$ homotopy equivalence

for $g_X \in \text{Met}_{K \equiv C}(X)$ can find (Yamabe!)
 $C \in \mathbb{R} \setminus \{0\}$

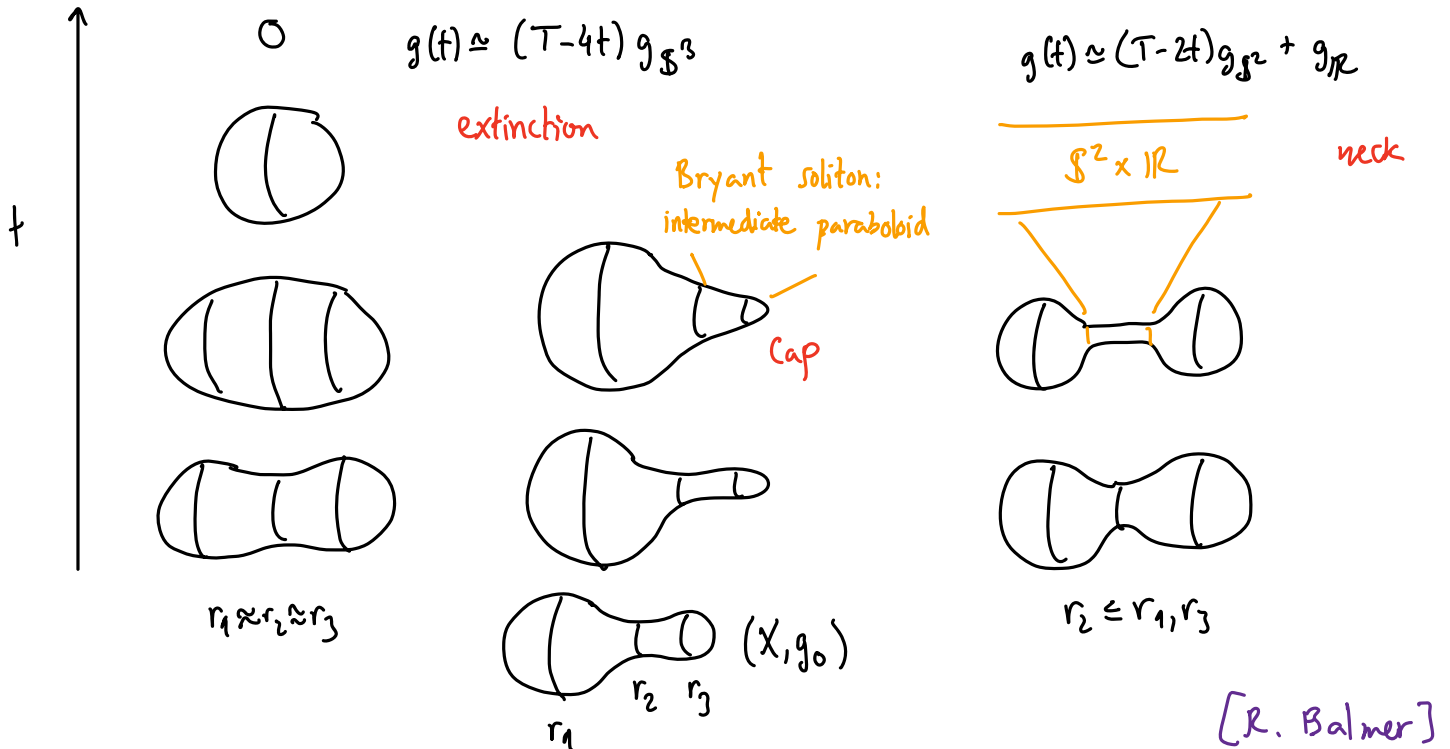
$\Leftrightarrow \text{Met}_{K \equiv C}(X)$ contractible



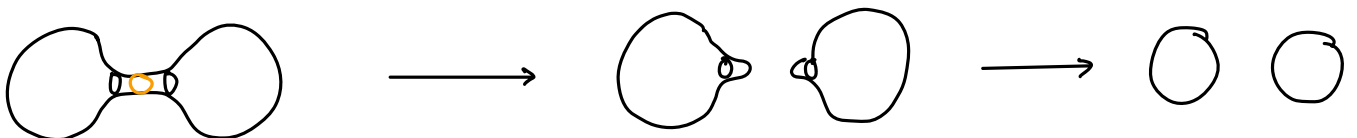
Ricci flow (for $X \neq S^3, \mathbb{RP}^3$)

2d blow-up: round metric becoming small

3d blow-up: more singular behaviors



Idea: combine with surgeries ("controlled cuts")



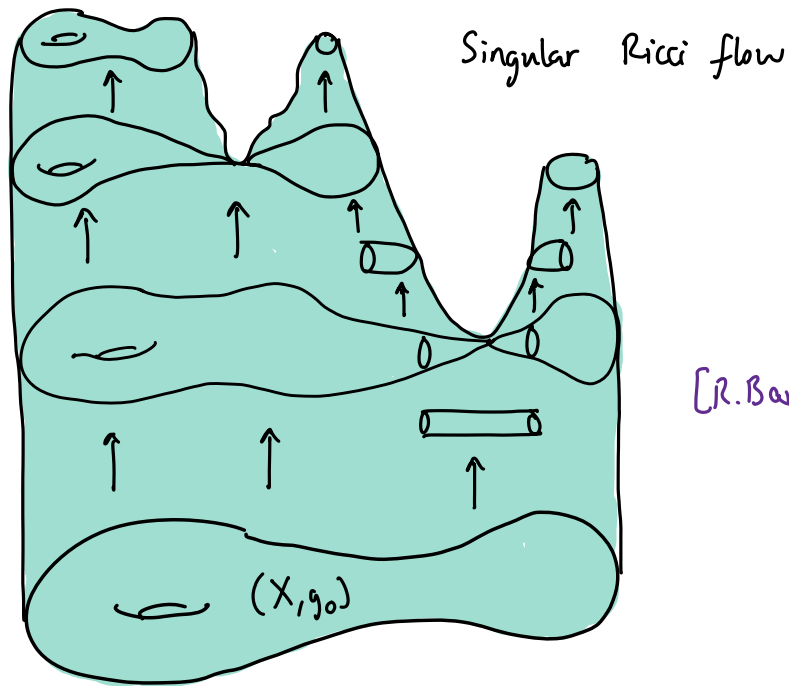
"take away a sphere just before running into a singularity"

Perelman: starting with a closed 3-manifold, this process always converges
 $\hookrightarrow \text{conn, opt, } \partial X = \emptyset$

Later: canonical way of performing the surgery (in limiting time)

$\rightarrow$ spacetime formulation (put together all time slices before & after surgeries)

- ⇒ [Thurston] Geometrization conjecture: "closed 3-mfds. can be decomposed in a canonical way in 8 types of pieces: 8 geometric building blocks in 3d"
- ⇒ Poincaré conjecture 3d: 1-conn., closed 3-mfd. is homeo. to S^3



[R. Bamler]